

APPENDIX C:

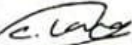
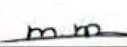
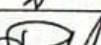
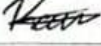
COLLABORATION DECLARATION STATEMENT

We hereby solemnly declare that the report is prepared and completed by us. We have collaborated equally in drawing up this report. This report has not been previously submitted to any other university, college or organization for an academic, qualification, certificate, diploma or degree. We each have signed and acknowledge the plagiarism declaration.

Project Title: Attendingly

Date: 19/09/2025

The **TEAM LEAD** student listed first in the table below is the only student to submit the assignment via Moodle. You will be allowed any number of submissions, but only the last submission will be marked. Penalty deductions will be made for multiple submissions by different student. An original signature (handwritten) is required – then scan or take a picture of this page and insert it as an image in your final project submission. Team size is 8 to 10 students.

	Student Number	Name	Signature
Team Lead	22310563	Avishandren Moodley	
2	22102825	Violan Reddy	
3	22235718	Kushal Maharaj	
4	22324091	Cacilda Langa	
5	22383672	marixole Mabhoza	
6	22382901	Atish Reddy	
7	22302289	Matthew JJC Rosa	
8	22330522	Prevani Moodley	
9	22332308	Raillin Arnachallam	
10	22208176	Andile Mahlaba	

(Minimum of 8 members and Maximum of 10.)



DURBAN UNIVERSITY OF TECHNOLOGY

Attendly: Revolutionising Classroom Attendance with Facial Recognition Technology

By
Group 5

Student No:	Name:	Surname:	Qualification Code:
22310563	Avishandren	Moodley	BINCT1
22332308	Railin	Arnachallam	BINCT1
22208776	Andile	Mahlaba	BINCT1
22302289	Matthew	Rosa	BINCT1
22382901	Atesh	Reddy	BINCT1
22330522	Prevani	Moodley	BINCT1
22324091	Cacilda	Langa	BINCT1
22383672	Masixole	Mabhoza	BINCT1
22102825	Violan	Reddy	ADICT1
22235718	Kushal	Maharaj	ADICT1

SFEN301

2025

Table of Contents

Process Model.....	4
1. Company and Problem Description	4
2. Process Model.....	5
Requirements Elicitation	5
1. User requirements	5
2. System Requirements	6
3. Functional Requirements	6
4. Non-functional Requirements.....	7
Project Management.....	7
1. Project Technology.....	7
2. Risks	8
3. Schedule/Timeline	9
4. Costs.....	9
Tools and Environment.....	10
1. Web Platform	10
2. Programming languages	10
3. Database	10
4. Operating Environment.....	10
5. IDE	10
Software Design	12
1. Flow Diagram	12
2. ERD	13
3. Sequence Diagram	13
3. Sequence Diagrams.....	14
4. Wireframe	15
Software Development	16
1. Backend.....	16
3. Database	17
4. GUI	18
Software Verification and Validation	19
1. Correct Input.....	19
2. Wrong Input	20
3. Testing Type.....	21
References	22

Process Model

Attendly: Revolutionising Classroom Attendance with Facial Recognition Technology

1. Company and Problem Description

Company Description

Educational institutions continually grapple with the significant issue of student absenteeism, which has challenges for academic achievement, operational effectiveness, and student persistence. Persistent absenteeism is a major factor behind elevated failure and dropout statistics, ultimately diminishing institutional performance and learner success. These challenges are frequently aggravated by outdated, manual attendance-tracking techniques, including physical sign-in sheets or oral roll calls, that prove not only inefficient and labour-intensive but also highly susceptible to inaccuracies and exploitation, such as impersonation or fraudulent check-ins.

In response to these systemic shortcomings, the university has prioritized the adoption of innovative technological solutions to cultivate a more accountable, data-informed educational atmosphere. Central to this strategy is the deployment of an automated platform that seamlessly integrates into daily academic routines, eliminating disruptions while vastly improving the reliability of participation records. This effort supports a larger mission to boost student involvement, enable precise attendance monitoring, and supply faculty and staff with trustworthy, up-to-date information to aid early intervention for vulnerable learners.

The envisioned platform, Attendly, is a web-enabled facial recognition attendance system crafted to fulfil these objectives. It streamlines the entire attendance process by enabling students to enrol their facial profiles through an intuitive online portal. Going forward, their presence is registered automatically, without the need for manual input, strengthening both precision and data authenticity. By integrating smoothly into existing academic workflows, Attendly reduces administrative overhead, frees educators to concentrate on instruction, and supplies the institution with precise, actionable intelligence.

Problem Description

A central obstacle to academic success lies in the continued use of obsolete and error-prone manual attendance systems, which actively undermine both instructional efficiency and student achievement. Conventional approaches, such as paper registers or verbal attendance calls, waste considerable class time on logistical chores and are easily subverted through dishonest acts like peer proxy sign-ins. These weaknesses produce attendance records that are neither accurate nor credible.

The repercussions of such unreliable data are profound. The integrity of academic assessment is compromised, as instructors cannot legitimately incorporate attendance into grading frameworks. At an administrative level, the inability to reliably detect prolonged or patterned absence means that students in danger of failing or withdrawing remain unnoticed until it is too late. Support measures, as a result, tend to be reactive rather than preventive.

Addressing these deficiencies demands an integrated solution that is automated, secure, and transparent. An effective system must fulfil several essential functions such as full automation, identity assurance, data security, and insightful reporting. The Attendly system is engineered to meet these requirements through a multifaceted security architecture. It incorporates facial authentication paired with IP-based location validation to ensure that students are physically present in the designated learning space. Furthermore, defined time constraints for clocking in and out promote punctuality and curb potential misuse. This comprehensive framework establishes a dependable and fraud-resistant protocol for recording student participation. With real-time logging and advanced reporting capabilities, Attendly enables instructional staff to incorporate attendance into evaluative metrics and helps institutional leaders identify at-risk students early, turning attendance data into a dynamic instrument for fostering academic accountability and success.

2. Process Model

The Attendly project is being developed using the Agile Software Development methodology, an approach characterized by iterative cycles, ongoing feedback, and the incremental release of working software. This framework is well-suited to the development of Attendly, as it supports adaptability to changing needs, continuous integration of user input, and the steady improvement of core features, including facial recognition, IP-driven location checks, and time-restricted attendance marking.

Work is organized into a sequence of sprints, each dedicated to a specific functional area. The first two sprints center on foundational activities: collecting system requirements, designing the architectural framework, and configuring the database. Subsequent sprints tackle the implementation of student registration, facial data enrolment, and the core mechanism for recording attendance. Later iterations introduce more advanced capabilities, including real-time attendance monitoring, clock-in/clock-out functions, and personalized dashboard interfaces for lecturers and administrators.

A key element of the process is the review phase following each sprint, during which stakeholders assess progress, test new features, and propose adjustments. This recurring cycle of build-and-evaluate ensures the system aligns with the university's standards for accuracy, security, and ease of use, while also providing opportunities to resolve technical challenges related to recognition algorithms, geolocation validation, and scalability.

By adopting Agile, the development team maintains a flexible, collaborative, and responsive workflow. This enables a sharp focus on high-priority requirements such as full automation, tamper-proof record keeping, and secure access controls. The result is a robust, reliable, and intuitive platform capable of tracking student attendance accurately and in real time.

Requirements Elicitation

1. User requirements

User requirements capture what different stakeholders, students, lecturers, and administrators, need from the system to solve the problem effectively. These requirements are high-level expectations that ensure the solution is practical, accessible, and addresses real-world challenges faced by its users.

Students

- Register facial data once and use it to mark attendance autonomously.
- Clock in and out within a lecturer-defined time window.
- Attendance is only valid if physically present in class (IP/location verification).
- Access the system via mobile phones, tablets, or computers (cross-platform).

Lecturers

- Real-time access to attendance records.
- Ability to generate reports and allocate marks.
- Receive notifications of irregular activity (late check-ins, unauthorized IPs).

Administrators

- Manage student and lecturer accounts securely.
- View historical data and generate institutional reports.
- Ensure data integrity through immutable records.

2. System Requirements

User requirements capture what different stakeholders, students, lecturers, and administrators, need from the system to solve the problem effectively. These requirements are high-level expectations that ensure the solution is practical, accessible, and addresses real-world challenges faced by its users.

Students

- Register facial data once and use it to mark attendance autonomously.
- Clock in and out within a lecturer-defined time window.
- Attendance is only valid if physically present in class (IP/location verification).
- Access the system via mobile phones, tablets, or computers (cross-platform).

Lecturers

- Real-time access to attendance records.
- Ability to generate reports and allocate marks.
- Receive notifications of irregular activity (late check-ins, unauthorized IPs).

Administrators

- Manage student and lecturer accounts securely.
- View historical data and generate institutional reports.
- Ensure data integrity through immutable records.

3. Functional Requirements

Functional requirements outline the specific actions and features the system must perform to achieve its purpose. These define “what the system should do,” ensuring that all essential operations, from logging in to generating attendance reports, are supported. The following are the functional requirements:

1. User Registration & Authentication

- Students register with institutional credentials and facial data.
- Role-based login for students, lecturers, and administrators.

2. Attendance via Facial Recognition

- Capture and verify the face in real time.
- Record attendance automatically once validated.

3. Time-Bound Attendance

- Enforce lecturer-defined clock-in/out windows.
- Reject late or repeated submissions.

4. Location Verification

- IP checks ensure attendance is classroom-only.

5. Attendance Records Management

- Securely log attendance with timestamps.
- Provide lecturers with reporting and exporting options.

6. Immutable Records

- Once marked, attendance cannot be altered or deleted.

7. Cross-Platform Access

- Accessible on web browsers (mobile and desktop).

4. Non-functional Requirements

Non-functional requirements describe how the system should perform rather than what it should do. They ensure the solution is reliable, secure, efficient, and user-friendly, which is critical for adoption and long-term sustainability.

Performance

- Facial recognition is processed within 3–5 seconds.
- Support hundreds of concurrent users.

Scalability

- Expandable to multiple faculties and campuses.
- Architecture must allow future feature integration (dashboards, analytics).

Security

- Encrypt facial data and login data (in transit and at rest).
- Strong authentication and authorization controls.
- Tamper-proof attendance records.

Usability

- Intuitive, mobile-first design.
- Minimal technical barriers for users.

Availability & Reliability

- $\geq 99\%$ uptime with redundancy.
- Backup and recovery protocols in case of system failure.

Maintainability

- Modular design for easier debugging and updates.
- Documentation for IT teams to ensure long-term support.

Project Management

1. Project Technology

The development of Attendly relies on a modern technology stack that balances performance, scalability, and ease of deployment. The front end is built using React.js, chosen for its component-based architecture, responsive design capabilities, and support for creating dynamic web applications that function seamlessly across devices, including smartphones and tablets [6][16]. React enables the team to design a user-friendly, mobile-first interface, which is essential since students must be able to clock in and out of attendance on their phones.

For the back end, Node.js is used due to its asynchronous, event-driven nature, which allows Attendly to handle many simultaneous attendance requests without latency. This ensures that the system remains responsive even during peak usage, such as the start of lectures when many students attempt to log attendance at once [12][14].

The database layer employs SupaBase, a NoSQL database that offers flexibility in storing complex data such as student profiles, facial recognition embeddings, and attendance records. SupaBase scalability and schema-less design makes it suitable for handling high volumes of attendance data while maintaining data integrity [7][18].

Facial recognition functionality is implemented using OpenCV in combination with deep learning frameworks, enabling robust face detection and verification under varying conditions. This ensures high recognition accuracy and reduces errors caused by lighting, angles, or facial occlusions [2][5].

The system is hosted on a local server currently but later will be implemented on a cloud platform s AWS or Microsoft Azure, providing scalability, reliability, and disaster recovery capabilities. This ensures that Attendly can scale to institutional requirements while maintaining availability and data security [10][15].

Together, this technology stack provides a secure, scalable, and high-performing environment for Attendly, ensuring that it meets the requirements of both users and institutions.

2. Risks

The implementation of Attendly involves several risks that need to be carefully managed to ensure successful deployment and adoption.

One of the primary risks is technical accuracy in facial recognition. Variations in lighting, camera quality, facial expressions, or the use of accessories such as masks and glasses may reduce recognition accuracy, leading to false positives or negatives [2][6]. Mitigating this risk requires the integration of advanced machine learning models and thorough testing in real classroom environments.

Another risk is data privacy and security. Since Attendly processes sensitive biometric data, there is a risk of data breaches or misuse of student information. Strong encryption, secure authentication, and compliance with data protection regulations such as POPIA in South Africa or GDPR internationally are essential to minimise this risk [5][10].

System misuse also poses a challenge. Students may attempt to spoof the system using photographs or videos to mark attendance fraudulently. To address this, Attendly must integrate anti-spoofing mechanisms such as liveness detection, which verifies that a live face is being presented [12][18].

Additionally, infrastructure risks may arise if the system experiences downtime during critical lecture periods. Limited internet connectivity or power failures on campus could prevent attendance from being recorded, potentially affecting student performance records. Cloud-based hosting with backup and redundancy can reduce this risk [7][14].

Finally, there are user adoption risks. Lecturers or students may resist using a new digital system due to unfamiliarity or concerns about privacy. This risk can be mitigated through training, awareness campaigns, and providing clear evidence of the system's reliability and fairness [1][15].

In summary, while Attendly faces risks in technical accuracy, security, misuse, infrastructure, and adoption, proactive strategies and robust design choices can effectively mitigate them, ensuring a reliable and trusted attendance solution.

3. Schedule/Timeline

The project is broken into the following phases:

Project Timeline (22 July – 29 September 2025):

- 1. Requirements Analysis and Planning (22–28 July)**
Define objectives, scope, and draft requirements after research on facial recognition, storage, and integration.
- 2. System Design (29 July – 8 August)**
Develop models, ERDs, process flows, wireframes, and plan the cloud database schema.
- 3. Development Phase I (11–22 August)**
Implement authentication, registration, login, and attendance modules; build the lecturer dashboard skeleton.
- 4. Development Phase II (25 August – 5 September)**
Integrate facial recognition with anti-spoofing, ensure mobile compatibility, and add reporting/export features.
- 5. Testing and Validation (8–12 September)**
Perform unit, integration, and acceptance testing, validate performance, and refine accuracy/security.
- 6. Documentation Finalisation (13–16 September)**
Compile project report and user manual; submit by 16 September.
- 7. Deployment and Presentation (17–29 September)**
Deploy prototype on the cloud, run final classroom testing, and deliver project presentation.

This timeline ensures that all deliverables, including documentation and system prototype, are completed on time while maintaining a balance between development quality and project deadlines.

4. Costs

The development and deployment of Attendly involve both one-time and recurring costs, which can be categorised into hardware, software, development, and operational expenses.

Hardware costs are minimal since the system leverages existing student and lecturer devices (computers and smartphones) and cloud infrastructure for hosting[6][12].

Software costs primarily relate to licences for development tools, cloud services, and libraries. React.js and Node.js are open source, reducing development costs. Cloud hosting (AWS, Azure, or Google Cloud) may incur recurring costs depending on storage, bandwidth, and compute usage, estimated at R20,000–R50,000 per month during peak academic periods [3][14].

Development costs include team labour, which encompasses front-end, back-end, and AI model development for facial recognition. Assuming a small team of three developers over approximately two months, development labour could be roughly R40,000–R60,000, depending on hourly rates [5][10].

Operational costs involve system maintenance, updates, security audits, and support. Monthly expenses for database backups, server management, and security monitoring may total R10,000–R30,000, ensuring that Attendly remains reliable and secure [1][15].

Indirect costs may include training lecturers and administrators on system usage, producing user manuals, and initial pilot testing. These are essential for smooth adoption and can be estimated at R5,000–R8,000.

In conclusion, while Attendly's deployment requires initial investment in development and minimal hardware, the system reduces long-term administrative costs by automating attendance management. Its scalable, cloud-based design ensures that recurring costs remain manageable while providing robust, secure, and efficient functionality [6][18].

Tools and Environment

1. Web Platform

Attendly is developed as a web-based application, ensuring accessibility across multiple devices, including desktops, laptops, and smartphones. This design eliminates the need for installation of native apps, allowing students and lecturers to access the system directly through a standard web browser. At present, the platform is hosted on a local development machine, which provides sufficient resources for testing, demonstrations, and controlled usage. While this setup is temporary, it ensures that the system remains fully functional and responsive during peak attendance simulations. Future deployment plans include migrating to a dedicated server or cloud-based environment to enhance scalability, availability, and long-term reliability. [6][16].

2. Programming languages

The system uses React.js for the front end, providing a responsive, component-based interface suitable for mobile-first design. Node.js powers the back end, enabling asynchronous request handling to support large numbers of simultaneous users. For facial recognition, Python is utilised with OpenCV and deep learning libraries, offering high accuracy in face detection and verification [2][5].

3. Database

Attendly's makes use of Supabase's real-time database which implements a PostgreSQL RDBMS (relational database management system). The choice for using a RDBMS over a NoSQL database is that a RDBMS has higher performance than a NoSQL database when dealing with formatted data and manipulating data to form tables using joins or other queries. As the database is being hosted by Supabase, the system makes use of API requests to perform various CRUD operations. The database is protected by RLS (row level security) policies which verifies if the user has the required role (admin, student or lecturer) to access the various tables to perform CRUD operations. Authorization is verified by the JWT (JSON Web Token) assigned to the user by Supabase.

4. Operating Environment

Attendly is designed to operate in a cross-platform environment, accessible via Windows, macOS, Linux, Android, and iOS devices. The system leverages web standards (HTML5, CSS3, and JavaScript) to ensure compatibility across browsers, while cloud hosting provides consistent uptime and reliability. This allows students to mark attendance from any device connected to the internet, while IP verification ensures that the location requirement is enforced [3][10].

5. IDE

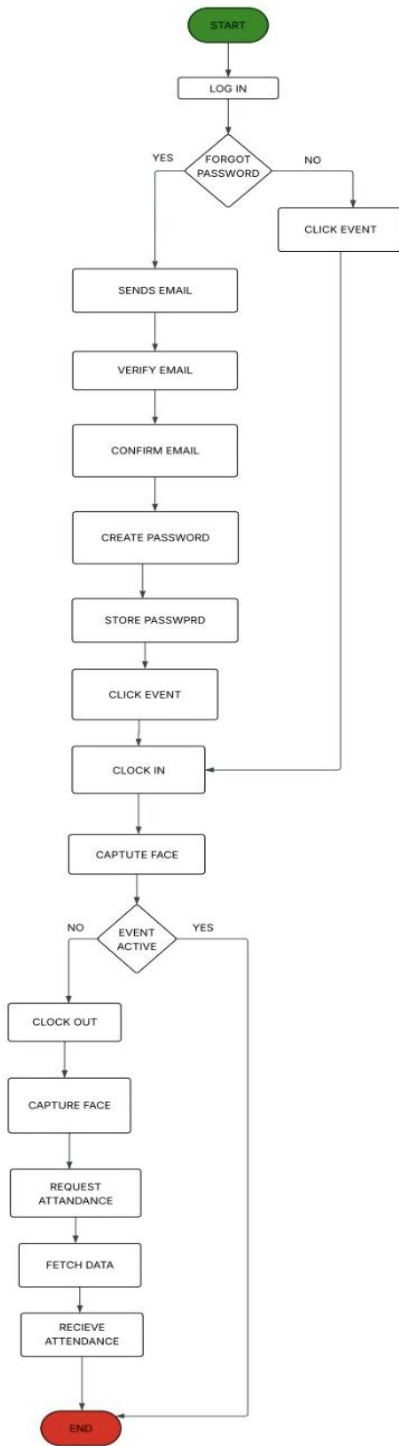
For development the chosen IDE for the project is Microsoft Visual Studio Code (VS Code). VS Code is the best choice for development on this project due to its ease of use, highly customizable, being an open-source IDE and its massive community. VS Code allows for easy development of all the chosen programming languages (JavaScript, Typescript, Python, HTML and CSS) and frameworks (VITE, Reactjs and Flask). VS Code's Extensions (plug-ins) are key to allow the effective debugging of code, refactoring, page viewing, unit testing and various other features. VS Code also offers extensions for GitHub (project repository) which integrate Github's workflows, making version control of branches easier to control and review, thus making it effective for collaboration and integrating various code snippets together. VS Code is highly customizable which allows for users to tailor their environment to

best tailor their needs. VS code highly active community is constantly releasing new plug-ins and improving on existing extensions which is a positive sign to a lasting development environment for this project and following projects improving the system.

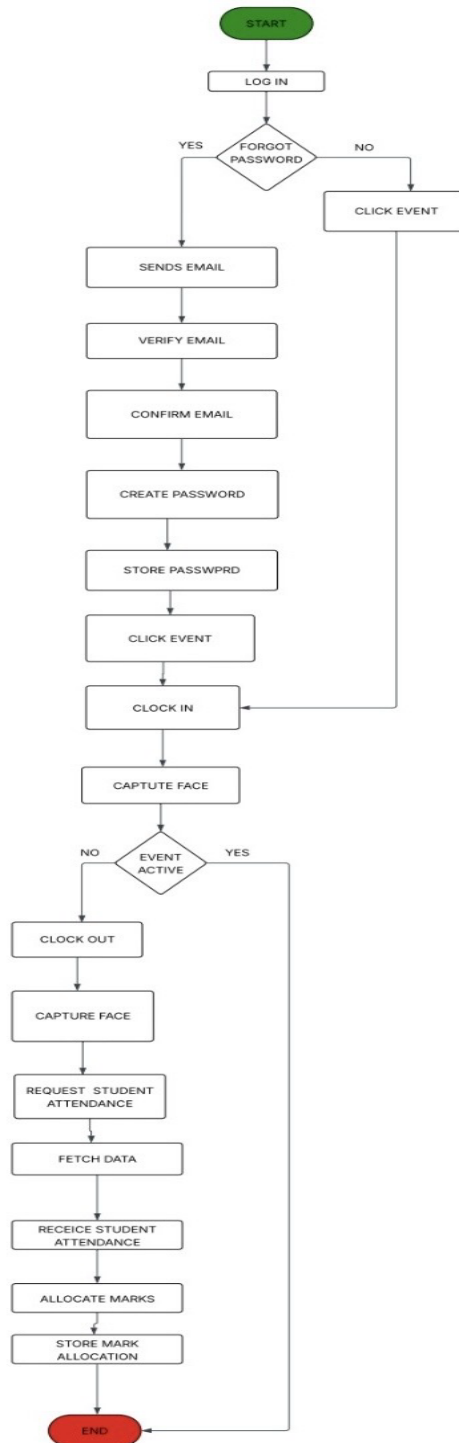
Software Design

1. Flow Diagram

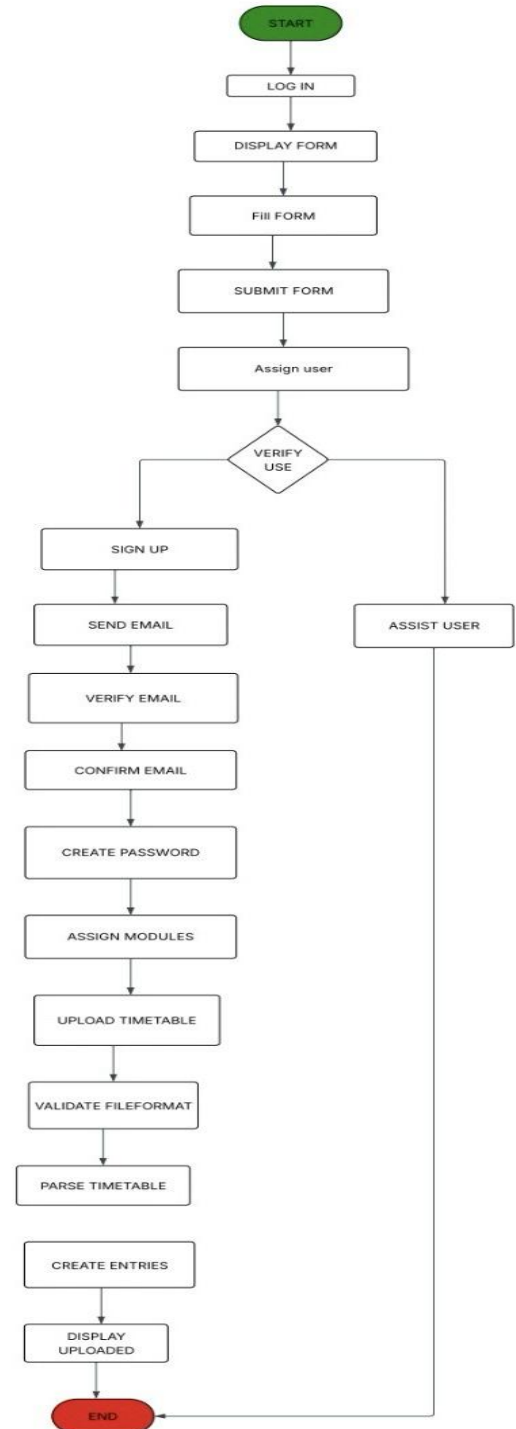
STUDENT:



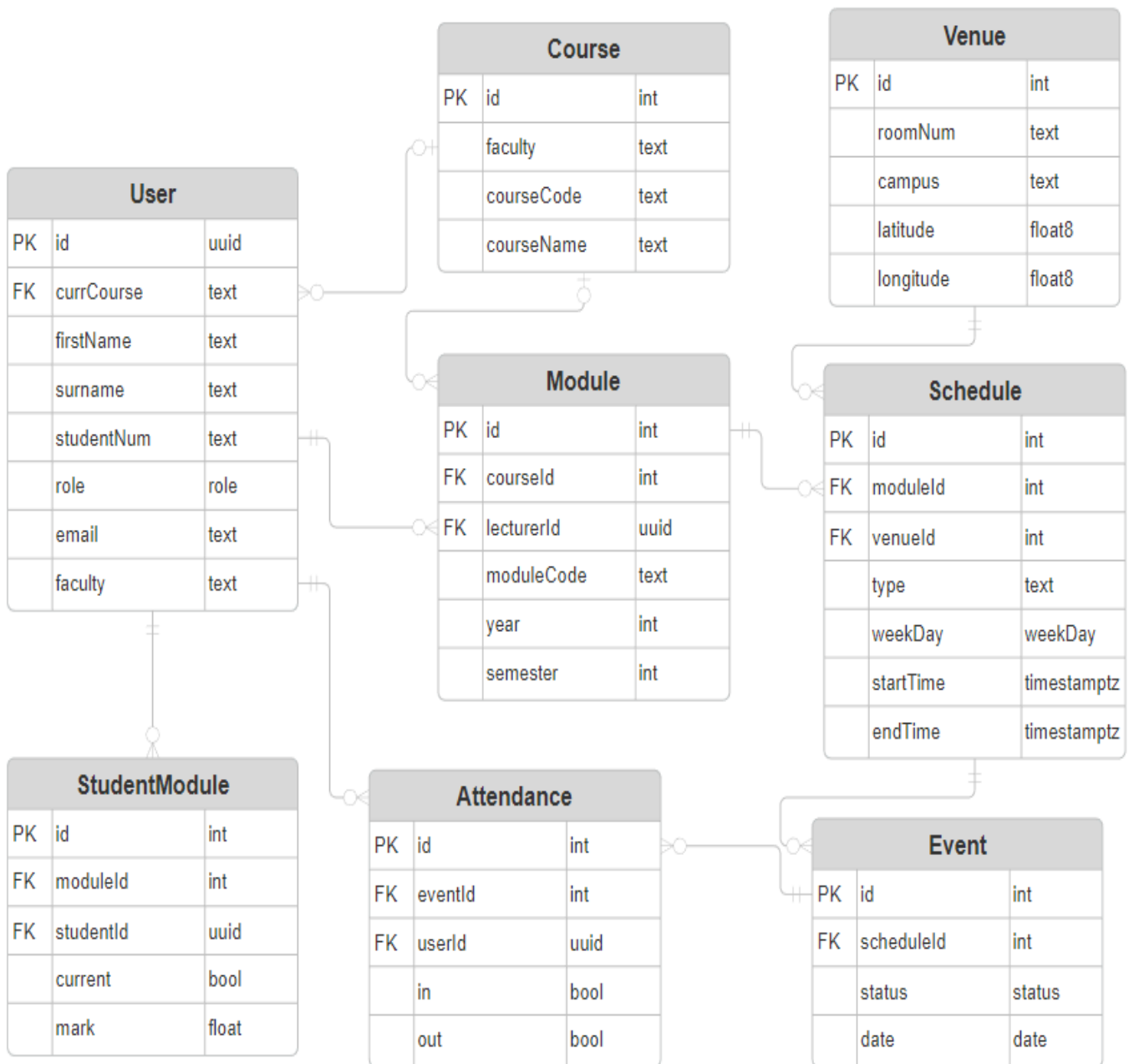
LECTURE:



ADMIN:



2. ERD



3. Sequence Diagrams

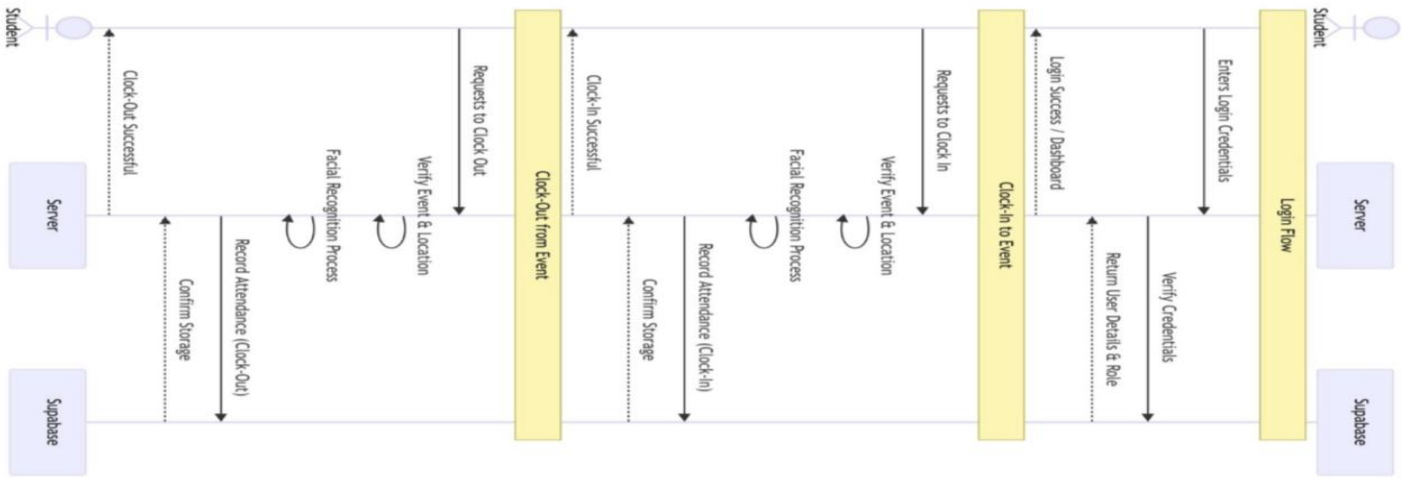


Figure 1. Student Sequence Diagram

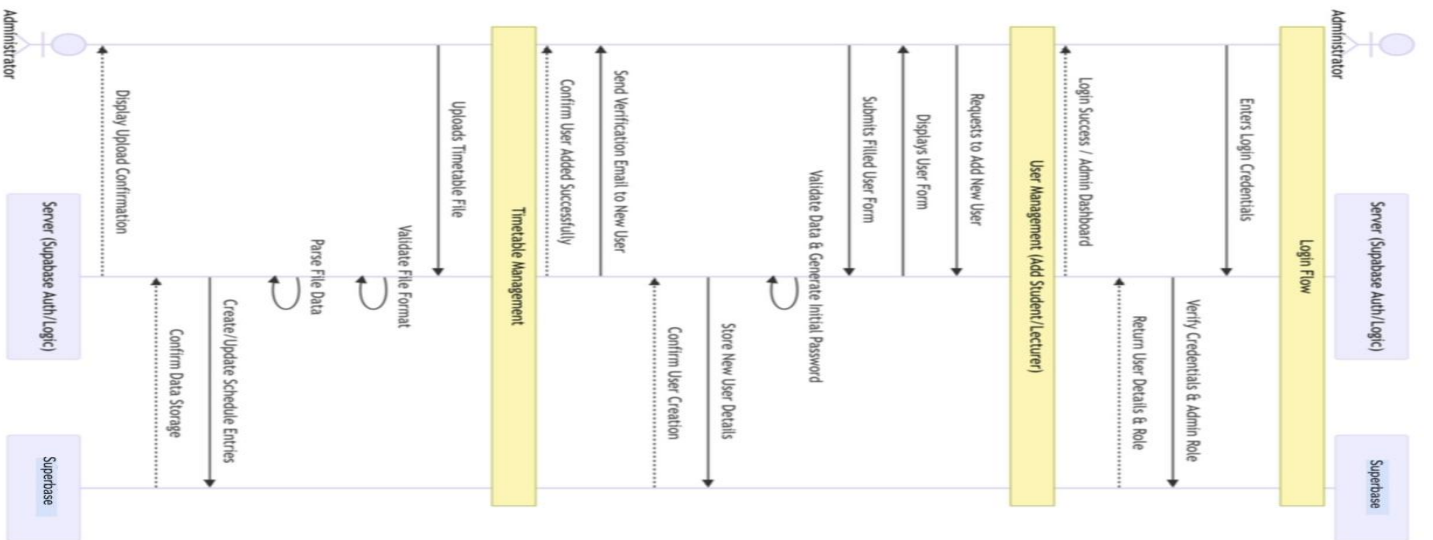


Figure 2. Admin Sequence Diagram

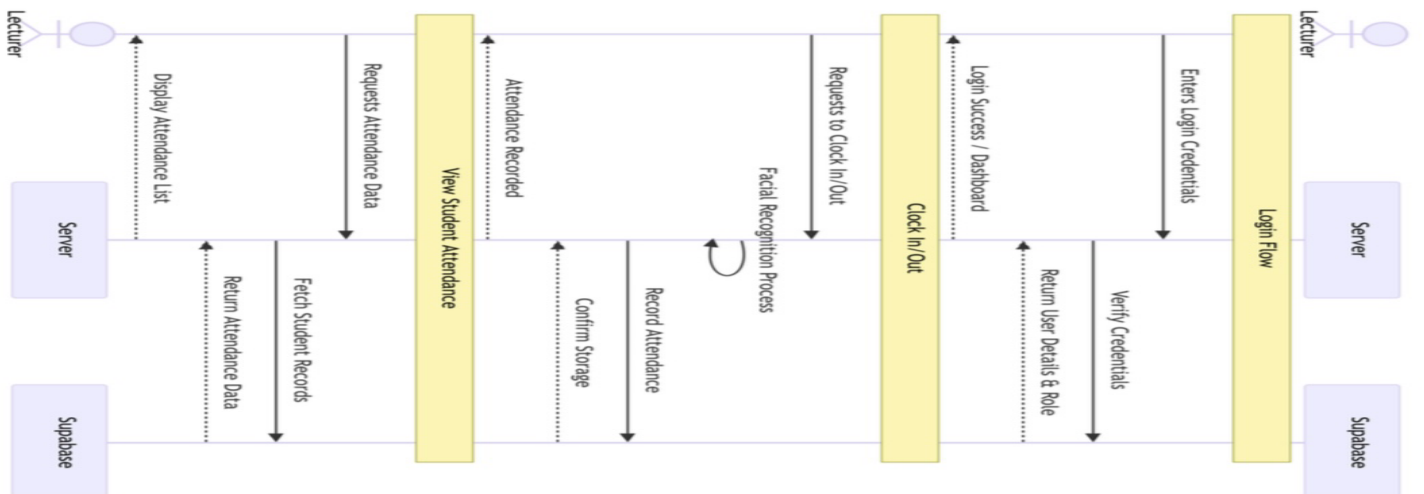


Figure 3. Lecturer Sequence Diagram

4. Wireframe

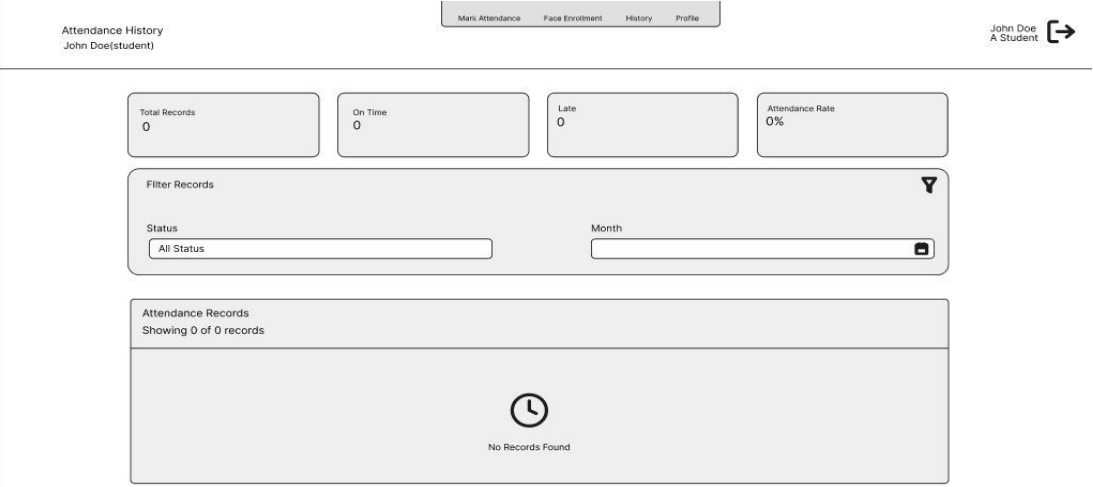


Figure 4. Student Dashboard

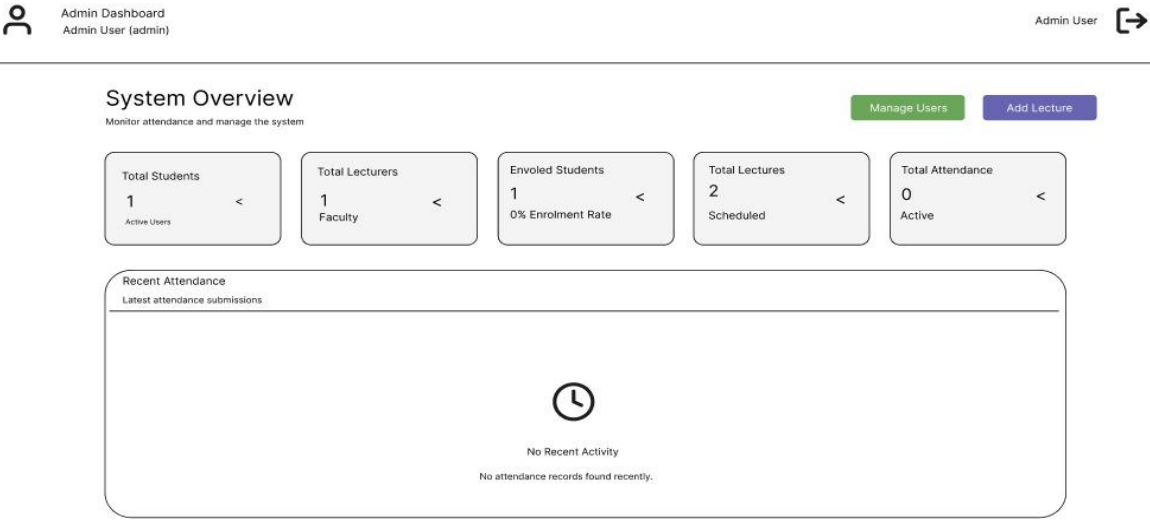


Figure 5. Admin Dashboard

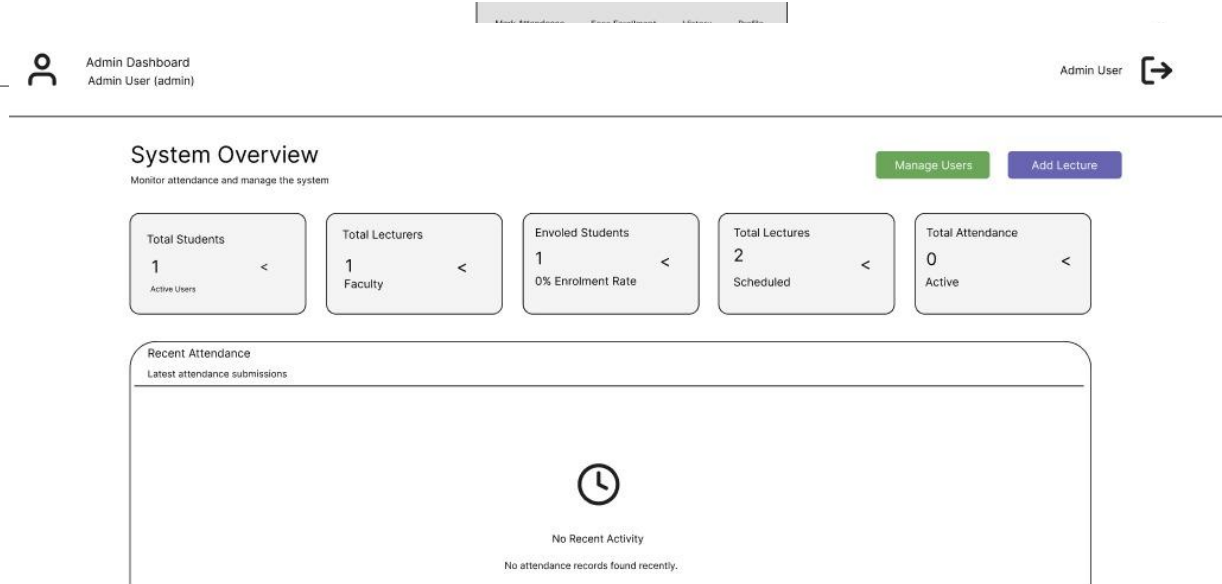


Figure 6. Student Portfolio

Software Development

1. Backend

The Attendly back end is the core component of the system, it is responsible for securely processing facial recognition , validating the user location, managing attendance records, allocating marks and allowing for new students to be added and old students to be removed. It ensures the integrity of the attendance process, protects sensitive student data, and allows for seamless interaction between all system components.

1. Key APIs:

- **Facial Recognition Authentication:** Provides an endpoint to which a student's image is submitted. The API returns an authentication result and, upon success, triggers the attendance allocation process.
- **Attendance & Marks Allocation:** This system automatically creates a new attendance record linked to the student, course, and timestamp. Upon successful attendance registration, it immediately allocates a mark for attendance in the student's record.
- **Data Retrieval for Dashboards:** Serves tailored data to each user role (Student, Lecturer, Admin) ensuring users only see information relevant to their permissions.

2. Authentication & Authorization:

- To secure the system, a combination of JWT (JSON Web Tokens) and flask_login were used for user authentication and session management. This ensures that all API requests are made by verified, logged-in users.
- **Login Process:** A user will login using their DUT credentials, when the request is submitted, the system will compare the provided information to the pre-existing data. If a match is found, the user will be signed into their account and taken to the allocated role's home page.
- **Role-Based Access Control (RBAC):** Distinct roles (Student, Lecturer, Admin) are assigned strict permissions. Students can only view their own attendance records and timetable. Lecturers can view attendance reports, statistics and marks for their courses. Admins have full system access to manage users and courses, though they too are unable to change attendance.
- **Facial Recognition & Geolocation Process**

Attendance is captured through a secure process that combines facial biometrics and location verification to prevent fraud and ensure accuracy and authenticity.

How the attendance process functions:

- **Image & Coordinate Submission:** A student uses the web application to capture their image and the device's GPS coordinates when attempting to mark attendance for a class.
- **Facial verification:** The back end receives the image and coordinates. It first checks the coordinates against the known venue location for the class. If valid, it then processes the image using a facial recognition algorithm to compare it against the verified photos stored in the database.
- **Update Attendance & Allocate Mark:** Upon successful verification of both face and location, the back end updates the attendance database, marking the student as 'Present' for that class

session. The system then automatically triggers a database operation to allocate the corresponding mark to the student's record for that specific attendance event.

3. Database

The database used to store and manage data is PostgreSQL, it is a relational database management system (RDMS). This database is ideal for our complex system with a large scale of data and with high performance for data retrieval and manipulation. The database for the attendance system is central to the system, which securely stores all stakeholder information such as student profiles, facial captures, attendance records, lecturer information, and course information. The key entities include User, venue, schedule, studentModule, Course, Auth, Attendance, Module, Event. They also have the relationships and attributes that go along with them and is used as the focus for designing the software system. The database is protected by authentication and authorization processes and access to the database is limited to protect all users' information. All users have unique login credentials to further secure information.

The database consists of the following entities:

- **User:** This entity stores information about all users, including full name, email address, student number, courses, and identification number.
- **Course:** This entity stores information abouts courses, including identification number, course code, faculty code, and course name.
- **Module:** This entity stores information about modules, including identification number, lecturer_id, course code, and module code
- **Student Module:** This entity stores information about student modules, including student id, module id, and marks
- **Schedule:** This entity stores information about lecture schedules including the module id, day of week, start time, end time, and venue.
- **Venue:** This entity stores information about the lecture venues, including the room number, campus and latitude and longitude coordinates.
- **Attendance:** This entity stores information about the attendance record, including user id, entrance time and exit time.
- **Event:** This entity stores information about a lecture, including the date and schedule.

The relationship between entities is as follow:

- Each student(user) is assigned to all registered modules(0 to many)
- Each student(user) is assigned to a registered course(0 to 1)
- Each student(user) is assigned to one login credential(1 to 1)
- A student (user) attendance is recorded for every lecture attended(0 to many)
- A degree/diploma(course) is assigned to students (0 to many)
- Every course has modules assigned to them(0 to many)
- Every module has lectures has a lecture(schedule) assigned to it(0 to many)
- Each module is assigned to a degree or diploma(0 to 1)
- Each module is assigned to a student(1 to 1)
- Each lecture(schedule) is assigned a venue to it(1 to 1)
- Lectures are assigned to modules(1 to 1)
- Each lecture is assigned to an event(0 to many)
- A lecture venue is assigned to the schedule(0 to many)
- Every attendance recorded is associated with a student(1 to 1)
- Every attendance recorded is assigned to an event(1 to 1)
- An event is associated with the schedule(1 to 1)
- An event is associated with every attendance recorded(0 to many)

4. GUI

The Graphical User Interface (GUI) of Attendly is designed to provide a clean, intuitive, and mobile-friendly experience for students and lecturers. Accessibility and ease of use are priorities to ensure smooth interaction with the system across all devices [6][16].

- **Login Page:**
 - **Entities:** Admin, Students, Lecturers
 - **Buttons:** Login, Forgot Password
- **Admin Board:**
 - **Features:**
 - Add users and user profiles
 - Upload timetables
 - Use CRUD functions on the database
 - View attendance analytics
 - Manage users
 - **Buttons:** Manage Users, Add Lecture
 - **Dashboard Widgets:** Total Students, Total Lectures, Enrolled Students, Total Lectures, Total Attendance, Recent Attendance
 - **Navigation:**
 - Manage Users
 - Add Lecture
 - Logout
- **Student Dashboard:**
 - **Features:**
 - View current and upcoming lectures
 - Clock-in and clock-out for attendance
 - Facial recognition to take attendance
 - View attendance records (current and previous semesters)
 - Password reset
 - **Buttons:** Capture Photo, Quick Actions
 - **Dashboard Widgets:** Upcoming Lectures, Account Security, Total Records, On Time, Late, Attendance Rate, Profile, Attendance Statistics, Recent Attendance
 - **Navigation:**
 - Mark Attendance
 - Face Enrolment
 - History
 - Profile
 - Logout
 - Quick Actions
- **Lecturer Dashboard:**
 - **Features:**
 - View current and upcoming lectures
 - Capture their own attendance
 - View attendance analytics (students per module)
 - Password reset
 - **Buttons:** Start Enrolment, Check-in Now, Start Enrolment
 - **Navigation:**
 - Check-in
 - Face Setup
 - History, Logout
 - **Dashboard Widgets:** Check-in to Lecture, Total Classes, Attendance Rate, Punctuality, On-Time Arrivals, Attendance Breakdown, Active Lectures

In summary the Attendly GUI ensures that students and lecturers can interact with the system efficiently, securely, and without confusion. By combining responsive design, clear feedback mechanisms, and intuitive navigation, the system promotes high adoption rates and smooth classroom operations.

Software Verification and Validation

1. Correct Input

Verification and validation of Attendly ensure that the system functions accurately and reliably when handling valid input data. Correct input refers to data provided by students and lecturers that meets the required format, constraints, and business rules. Proper handling of correct input guarantees accurate attendance records, secure authentication, and reliable reporting [2][6].

Student Login and Registration:

- **Valid Email Format:** The system only accepts emails that match the university's official format (e.g., 00000000@dut4life.ac.za).
- **Password Constraints:** Passwords must meet minimum security requirements (e.g., 8 characters, including letters and numbers).
- **Face Registration:** Students must provide a clear facial image during registration. The system verifies image quality and ensures the face is correctly detected before generating the face embedding.
- **Expected Outcome:** Correct input leads to successful registration or login, and the student can access their dashboard and active sessions [5][10].

Attendance Submission:

- **Session Selection:** Students select an active session within the allowed time window.
- **Face Capture:** The captured image matches the stored face embedding with high confidence.
- **IP Verification:** Student device is within the authorised IP range, confirming presence in the classroom.
- **Clock-In/Clock-Out:** Attendance is successfully recorded in the database with timestamp, session ID, and status "Present."
- **Expected Outcome:** Students see an immediate confirmation message, and lecturers can view updated attendance reports.

Lecturer Input:

- **Course and Session Details:** Lecturers provide valid course ID, session date, start and end time, and optional IP restrictions.
- **Participation Marks:** Marks entered must fall within a valid range (e.g., 0–100).
- **Expected Outcome:** The system creates valid attendance sessions, stores session details securely, and displays reports correctly.

Testing for Correct Input:

- **Unit Tests:** Validate functions such as email verification, password hashing, and facial recognition.
- **Integration Tests:** Ensure that data flows correctly from frontend to backend and then to the database.

- **End-to-End Tests:** Simulate student attendance submissions and lecturer session creation to verify the entire process functions as expected [6][14].

In summary by rigorously testing for correct input, Attendly ensures that student attendance is accurately captured, stored, and reported, while maintaining data integrity and system security. This step is critical for building confidence in the system's reliability and usability.

2. Wrong Input

Attendly must handle wrong or invalid input gracefully to prevent errors, maintain security, and ensure data integrity. Wrong input refers to any data that does not meet system requirements, such as incorrect email formats, mismatched facial data, or attempts to manipulate attendance. Proper handling of wrong input ensures that students cannot fraudulently record attendance and that lecturers' data remains accurate [2][6].

Student Login and Registration Errors:

- **Invalid Email Format:** If a student enters an email outside the university format (e.g., 00000000@dut4life.ac.za), the system rejects the input with a clear error message.
- **Weak or Missing Passwords:** The system prevents registration if the password does not meet security criteria.
- **Duplicate Registration:** Students attempting to register multiple times with the same email are blocked, and the system notifies them of existing accounts.
- **Face Capture Errors:** Blurred, partially visible, or obstructed facial images are rejected. Students are prompted to retry with proper positioning and lighting.

Attendance Submission Errors:

- **Time Window Violation:** If a student attempts clock-in outside the allowed session time, attendance is denied, and a notification explains the restriction.
- **IP Mismatch:** Attendance requests from devices outside the authorised classroom network are rejected to prevent remote fraud.
- **Face Mismatch:** If the captured face does not match the stored face embedding, the system denies attendance and prompts a retry.
- **Repeated Clock-Ins:** The system prevents multiple submissions for the same session to maintain record integrity [3][12].

Lecturer Input Errors:

- **Invalid Course or Session Data:** The system checks that courses exist and that session dates and times are valid before creation.
- **Incorrect Participation Marks:** Marks outside the accepted range (e.g., negative values or above 100) are rejected, with a prompt to correct the input.

Error Handling Mechanisms:

- **Real-Time Validation:** Forms check inputs before submission to reduce errors.
- **Server-Side Validation:** Backend confirms data integrity and security, preventing tampering or bypassing of frontend checks.
- **User Feedback:** Clear error messages guide students and lecturers to correct input issues quickly.

- **Logging:** All failed attempts are logged for auditing purposes, helping administrators detect repeated misuse or potential fraud [6][14].

In summary Attendly's handling of wrong input ensures that attendance data remains accurate and secure, preventing manipulation or errors from invalid submissions. By combining real-time validation, server-side checks, and informative feedback, the system maintains reliability while guiding users toward correct usage.

3. Testing Type

To ensure Attendly is reliable, secure, and performs as expected, a combination of software testing types is employed. Testing covers both functional and non-functional aspects, verifying that students' attendance is accurately recorded, data integrity is maintained, and system performance meets university requirements [2][6][12].

1. Unit Testing:

- **Purpose:** Verify individual components, such as login, registration, attendance capture, and facial recognition algorithms.
- **Example Tests:**
 - Email and password validation
 - Face embedding generation
 - IP verification during clock-in
- **Tools:** Jest and Mocha for backend Node.js functions.

2. Integration Testing:

- **Purpose:** Ensure that different system modules interact correctly.
- **Example Tests:**
 - Backend APIs correctly store and retrieve attendance records from the database.
 - Attendance capture integrates correctly with the facial recognition microservice.
 - Lecturer dashboard accurately displays session data and reports [3][14].

3. System Testing:

- **Purpose:** Evaluate the complete Attendly system in an environment like production.
- **Example Tests:**
 - Students successfully clock in/out within allowed time windows.
 - IP/location restrictions are enforced.
 - Attendance reports reflect real-time data.
- **Performance Metrics:** Response times, concurrency handling, and error rates are measured to ensure reliability.

4. User Acceptance Testing (UAT):

- **Purpose:** Validate the system meets end-user requirements and is intuitive to use.
- **Example Activities:**
 - Students perform registration, clock-in, and view attendance history.

- Lecturers create sessions, view reports, and assign participation marks.
- **Feedback:** UAT identifies usability issues, helping refine interface elements and workflows.

5. Regression Testing:

- **Purpose:** Ensure that system updates or fixes do not introduce new errors.
- **Example Tests:** Re-run existing unit, integration, and system tests after updates to verify consistent performance.

6. Security and Penetration Testing:

- **Purpose:** Identify vulnerabilities that could compromise sensitive data such as facial embeddings, student information, or session records.
- **Example Tests:**
 - Attempt SQL/NoSQL injections on database APIs
 - Test access control for student and lecturer roles
 - Validate encrypted transmission of sensitive data (HTTPS, JWT) [6][18]

In summary Attendly employs a comprehensive testing strategy combining unit, integration, system, user acceptance, regression, and security testing. This ensures that the system is robust, reliable, and secure, capable of accurately capturing attendance while preventing fraud or data tampering, providing confidence for both students and lecturers.

References

- [1] E. Indra, A. F. Ramdhani, H. Wicaksono, M. H. Arifin and M. R. Siregar, "Design and Implementation of Student Attendance System Based on Face Recognition by Haar-Like Features Methods," 2020 3rd International Conference on Mechanical, Electronics, Computer, and Industrial Technology (MECnIT), Medan, Indonesia, 2020, pp. 336-342, doi: 10.1109/MECnIT48290.2020.9166595.
- [2] N. Narkhede, A. Menon, I. Mathane, S. Nikam and S. Dange, "Facial Recognition and Machine Learning-based Student Attendance Monitoring System," 2023 3rd International Conference on Intelligent Technologies (CONIT), Hubli, India, 2023, pp. 1-7, doi: 10.1109/CONIT59222.2023.10205631.
- [3] A. Ghodekar, T. Gunjal, A. Landage, A. Phapale, D. Thorat and R. Nikam, "A Face Recognition-Based Smart Student Attendance and Activeness Monitoring System," 2023 4th International Conference on Computation, Automation and Knowledge Management (ICCAKM), Dubai, United Arab Emirates, 2023, pp. 1-5, doi: 10.1109/ICCAKM58659.2023.10449572.
- [4] J. F. Rusdi, F. R. Kodong, R. E. Indrajit, H. Sofyan, A. Abdurrohman and R. Marco, "Student Attendance using Face Recognition Technology," 2020 2nd International Conference on Cybernetics and Intelligent System (ICORIS), Manado, Indonesia, 2020, pp. 1-4, doi: 10.1109/ICORIS50180.2020.9320819.
- [5] M. Abuzar, A. b. Ahmad and A. A. b. Ahmad, "A Survey on Student Attendance System Using Face Recognition," 2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions) (ICRITO), Noida, India, 2020, pp. 1252-1257, doi: 10.1109/ICRITO48877.2020.9197815.

- [6] M. A. B. M. Latif, R. A. Aziz and H. Hafit, "Development of Student Face Recognition Attendance System," 2024 IEEE 22nd Student Conference on Research and Development (SCOReD), Selangor, Malaysia, 2024, pp. 134-139, doi: 10.1109/SCOReD64708.2024.10872759.
- [7] S. -H. Lee, M. -K. Sohn, H. Kim and J. Kim, "Design Approach of Electronic Attendance-Absence Recording System using Multi-User Face Recognition," 2023 Congress in Computer Science, Computer Engineering, & Applied Computing (CSCE), Las Vegas, NV, USA, 2023, pp. 2773-2774, doi: 10.1109/CSCE60160.2023.00460.
- [8] K. Tapyou, P. Chaisil and J. Muangprathub, "Smart School Attendance System using Face Recognition with Near Optimal Imaging," 2021 18th International Joint Conference on Computer Science and Software Engineering (JCSSE), Lampang, Thailand, 2021, pp. 1-5, doi: 10.1109/JCSSE53117.2021.9493844.
- [9] R. Samet and M. Tanriverdi, "Face Recognition-Based Mobile Automatic Classroom Attendance Management System," 2017 International Conference on Cyberworlds (CW), Chester, UK, 2017, pp. 253-256, doi: 10.1109/CW.2017.34.
- [10] S. S. Rajawat and K. Saxena, "Face Recognition based Attendance System," 2022 1st IEEE International Conference on Industrial Electronics: Developments & Applications (ICIDeA), Bhubaneswar, India, 2022, pp. 95-99, doi: 10.1109/ICIDeA53933.2022.9970084.
- [11] S. Salvi and J. S. A. Pramod, "ClassScan: Simplifying Classroom Attendance using Three-Dimensional Dense Face Alignment and Face Recognition," 2023 7th International Conference on Computation System and Information Technology for Sustainable Solutions (CSITSS), Bangalore, India, 2023, pp. 1-6, doi: 10.1109/CSITSS60515.2023.10334242.
- [12] V. A, R. R. Krishna and R. V. U, "Facial Recognition System for Automatic Attendance Tracking Using an Ensemble of Deep-Learning Techniques," 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kharagpur, India, 2021, pp. 1-6, doi: 10.1109/ICCCNT51525.2021.9580098.
- [13] D. Yadav, S. Maniar, K. Sukhani and K. Devadkar, "In-Browser Attendance System using Face Recognition and Serverless Edge Computing," 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT), Kharagpur, India, 2021, pp. 01-06, doi: 10.1109/ICCCNT51525.2021.9580042.
- [14] M. Joshi, S. Sikka, M. Pandey and S. Tripathi, "Advance Attendance Management Through Face Recognition," 2025 7th International Conference on Signal Processing, Computing and Control (ISPCC), Solan, India, 2025, pp. 954-959, doi: 10.1109/ISPCC66872.2025.11039556.
- [15] S. K. Abirami, S. Jyothikamalesh, M. Sowmiya, S. Abirami, S. Angel Latha Mary and C. Jayasudha, "AI-based Attendance Tracking System using Real-Time Facial Recognition," 2022 6th International Conference on Electronics, Communication and Aerospace Technology, Coimbatore, India, 2022, pp. 1330-1335, doi: 10.1109/ICECA55336.2022.10009331.
- [16] P. Netinant, N. Akkharasup-Anan and M. Rakhiran, "Class Attendance System using Unimodal Face Recognition System based on Internet of Educational Things," 2023 IEEE 6th Eurasian Conference on Educational Innovation (ECEI), Singapore, Singapore, 2023, pp. 67-70, doi: 10.1109/ECEI57668.2023.10105374.
- [17] S. Rugtanom, J. Arunruerk, W. Yaemvachi, P. Sunantapot and P. Srikram, "Web application and Mobile application based Student Attendance Management System for Facial Recognition Attendance," 2024 23rd International Symposium on Communications and Information Technologies (ISCIT), Bangkok, Thailand, 2024, pp. 181-185, doi: 10.1109/ISCIT63075.2024.10793682.

- [18] A. M and S. V, "Face Recognition Based Attendance Management System Using Deep Learning Method," 2024 Second International Conference on Emerging Trends in Information Technology and Engineering (ICETITE), Vellore, India, 2024, pp. 1-7, doi: 10.1109/ic-ETITE58242.2024.10508036.
- [19] N. K. Jayant and S. Borra, "Attendance management system using hybrid face recognition techniques," 2016 Conference on Advances in Signal Processing (CASP), Pune, India, 2016, pp. 412-417, doi: 10.1109/CASP.2016.7746206.
- [20] B. Handaga, B. Murtiyasa and J. Wantoro, "Attendance System based on Deep Learning Face Recognition without Queue," 2019 Fourth International Conference on Informatics and Computing (ICIC), Semarang, Indonesia, 2019, pp. 1-4, doi: 10.1109/ICIC47613.2019.8985697.
- [21] A. D. Alexander, R. Salkiawati, H. Lubis, F. Rahman, H. Herlawati and R. T. Handayanto, "Local Binary Pattern Histogram for Face Recognition in Student Attendance System," 2020 3rd International Conference on Computer and Informatics Engineering (IC2IE), Yogyakarta, Indonesia, 2020, pp. 152-156, doi: 10.1109/IC2IE50715.2020.9274621.
- [22] M. Kamruzzaman, M. E. A. Sheikh, M. R. Hossain and M. M. B. Mojumdar, "Automated Attendance system for the Entire Classroom with Multi-Face Facial Recognition via AWS Rekognition," 2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), Chittagong, Bangladesh, 2025, pp. 1-6, doi: 10.1109/ECCE64574.2025.11013462.
- [23] C. Anilkumar, B. Venkatesh and S. Annapoorna, "Smart Attendance System with Face Recognition using OpenCV," 2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), Trichy, India, 2023, pp. 1149-1155, doi: 10.1109/ICAISS58487.2023.10250715.
- [24] A. Y. Strueva and E. V. Ivanova, "Student Attendance Control System with Face Recognition Based on Neural Network," 2021 International Russian Automation Conference (RusAutoCon), Sochi, Russian Federation, 2021, pp. 929-933, doi: 10.1109/RusAutoCon52004.2021.9537386.
- [25] A. Thakur, A. Pande, N. Pande, R. Khandelwal and P. Dwibedy, "Revolutionizing Classroom Engagement with Face Recognition and Attention-Based Attendance," 2025 3rd International Conference on Disruptive Technologies (ICDT), Greater Noida, India, 2025, pp. 1373-1377, doi: 10.1109/ICDT63985.2025.10986679.